

EVENT-DRIVEN CSV DATA INGESTION AND GRAPH-BASED CHATBOT

Development Process, Honest Reflection, and Project Journey

How the system was built, the challenges faced, and lessons learned

Designed and implemented a complete event-driven data ingestion system, focused on automation, scalability, and ease of use, and built using modern backend and graph database technologies.

Development Process

- Analyzed the problem statement.
- Designed system architecture.
- Selected appropriate technologies.
- Developed frontend.
- Built backend APIs.
- Integrated Kafka.
- Connected Neo4j.
- Implemented chatbot.
- Tested complete workflow.
- Prepared documentation.

Project Workflow

- Upload CSV.
- Validate dataset.
- Clean data.
- Send rows to Kafka.
- Consume messages.
- Store data in Neo4j.
- Receive user question.
- Generate Cypher query.
- Retrieve answer.
- Display chatbot response.

Challenges Faced

- Kafka configuration.
- Neo4j connection.
- CSV validation.
- Graph modeling.
- Duplicate prevention.
- API integration.
- Docker networking.
- End-to-end testing.

Solutions Implemented

- Modular architecture.
- Producer-Consumer model.
- MERGE for duplicate prevention.
- Dataset validation.
- Data cleaning pipeline.
- REST API communication.
- Docker Compose deployment.

Lessons Learned

- Event-Driven Architecture.
- REST API development.
- Kafka messaging.
- Graph databases.
- Cypher query language.
- Docker containerization.
- System integration.

Honest Reflection

Successfully Completed

- CSV upload.
- Dataset cleaning.
- Kafka integration.
- Neo4j integration.
- Graph storage.
- Chatbot.
- Cypher querying.
- Docker deployment.

Areas for Improvement

- Advanced NLP.
- Authentication.
- Cloud deployment.
- Performance optimization.
- Graph visualization.
- Multi-file support.

Future Roadmap

- AI-powered natural language understanding.
- Cloud-native deployment.
- Real-time analytics.
- Interactive dashboards.
- PDF and Excel ingestion.
- User authentication.
- Role-based access.
- Monitoring and logging.

Final Conclusion

Successfully built an event-driven CSV ingestion platform that integrates FastAPI, Kafka, Neo4j, and Cypher into a modular architecture. The project demonstrates automated data ingestion, graph-based storage, and conversational querying, delivering a scalable foundation that can be extended with advanced AI and enterprise features.

One-Line Project Summary

An event-driven, graph-based data ingestion platform that automates CSV processing using Apache Kafka, stores structured information in Neo4j, and enables natural language querying through a FastAPI-powered chatbot.

Key Technical Keywords

- Event-Driven Architecture
- Data Ingestion Pipeline
- REST API
- FastAPI
- Apache Kafka
- Producer
- Consumer
- Message Broker
- Asynchronous Processing
- Neo4j Graph Database
- Graph Nodes
- Relationships
- Cypher Query Language
- Pandas
- Dataset Validation
- Dataset Cleaning
- MERGE Operation
- Grounded Chatbot
- Docker

- Docker Compose
- Modular Architecture
- Scalable System
- Dynamic CSV Upload